Energy Sustainability in Data Centers: A Case Study of the Open Compute Project

Raab Iane

Institute of Technology and Leadership

Module 3

Professor Filipe Gonçalves

April 10, 2026

Abstract

This study investigated how data centers can be developed and operated in an environmentally sustainable manner, using the Open Compute Project (OCP) as a reference. A descriptive single-case study method was adopted, conducted according to the methodology of Robert K. Yin (1998), which consists of a structured approach to investigating contemporary phenomena in depth within their real-life context, using multiple sources of evidence and systematic analytical procedures, with evidence collected from academic publications, independent reports, industry technical reports, and official OCP documentation, organized into a hierarchy of reliability to control bias through triangulation, that is, the strategy of cross-checking and comparing different data sources to reduce distortions and enhance the validity of the findings. Results were structured across four layers of systemic intervention: (1) reconfiguration of the power distribution architecture, with efficiency gains from higher voltage (48 V DC) adoption and rack-level power shelf centralization; (2) functionality-oriented (vanity-free) design, with documented 15–20% reductions in energy consumption through aerodynamic and geometric rack redesign; (3) open-source energy management software (OpenBMC/OpenRMC), enabling granular monitoring and stranded power control; and (4) circularity and lifecycle design strategies addressing Scope 3 emissions, estimated at 38–69% of total carbon footprint. As an original contribution, the research identified that PUE (Power Usage Effectiveness), the sector's dominant metric, captures only approximately 30% of total environmental impacts and is structurally insufficient for guiding sustainability decisions in data centers. Findings indicate that energy sustainability requires systemic articulation of technical innovation, open standardization, and transparent governance.

Keywords: data centers; energy sustainability; Open Compute Project; Green AI; energy efficiency; Power Usage Effectiveness; life cycle.

Introduction

The accelerated growth of artificial intelligence (AI) in recent decades has generated unprecedented energy demand on global digital infrastructure. Data centers — physical structures responsible for processing, storing, and distributing data on a large scale — constitute the main point of concentration of this consumption. According to the International Energy Agency (IEA, 2023), in 2022 data centers consumed between 240 and 340 TWh, corresponding to about 1% to 1.5% of the electricity generated worldwide. More recent projections indicate that this consumption should double by 2030, driven especially by the expansion of AI workloads, which are growing approximately 15% per year — more than four times higher than other sectors (IEA, 2025).

Given this scenario, the question guiding this research is: how can data centers be developed and implemented in an environmentally sustainable way, considering the high energy consumption required by this technology? This question is relevant from both a practical and theoretical point of view, as it challenges the traditional criteria for evaluating computer systems, historically focused on performance and cost, and not on environmental efficiency.

To investigate this issue, this work adopts as its empirical object the Open Compute Project (OCP), a non-profit foundation that develops and disseminates open hardware specifications for data centers. The OCP represents a unique case in the sector: instead of protecting proprietary innovations, the foundation publishes its technical specifications openly, allowing any data center operator to adopt, adapt, and improve the solutions developed collaboratively by its members. This model establishes a bridge between technical innovation

and systemic dissemination, making the OCP a relevant object of study for understanding how sustainable practices can be created and scaled.

The study is based on the framework of Green AI (Ferraz, 2024), which proposes that the development of artificial intelligence systems should incorporate energy efficiency, transparency, and environmental responsibility as core performance dimensions — rather than as secondary considerations to accuracy metrics. From this perspective, the research analyzes OCP's practices across four layers of systemic intervention, examining not only their technical foundations but also their dissemination mechanisms, limitations, and implications for sustainability governance in data centers.

The specific objectives of this study are: (a) to identify and describe the main energy sustainability strategies adopted in the context of OCP; (b) to analyze how the interaction between design principles, operational practices, and standardization mechanisms influences energy efficiency on a global scale; (c) to identify challenges and limitations in the implementation of these practices; and (d) to evaluate the adequacy of the energy performance metrics currently used to guide sustainability decisions in data centers.

Literature Review and Theoretical Framework

Energy Consumption of Artificial Intelligence

The relationship between AI and energy consumption has gained academic attention through pioneering studies that measured the computational cost of training deep learning models. Strubell et al. (2019) demonstrated that training large-scale natural language models — AI systems designed to understand and generate human text — can produce carbon emissions

equivalent to those of five cars over their lifetime. Luccioni et al. (2022), in estimating the carbon footprint of the BLOOM model with 176 billion parameters, highlighted that while emissions from training are significant, continuous inference, due to its temporal and volumetric scale, can result in an even greater cumulative impact over the lifecycle of these systems.

Wu et al. (2021) advanced the analysis by proposing a systemic approach to AI sustainability, highlighting that the full lifecycle of systems includes data collection, training, inference, and hardware disposal, and that efficiency should be treated as a performance criterion across all these phases. Schneider et al. (2025), in a lifecycle analysis of AI hardware, demonstrated that equipment manufacturing accounts for approximately 25% of total infrastructure emissions over its lifespan, emphasizing the relevance of embodied energy as a key dimension of environmental footprint.

The International Energy Agency (IEA) (2023) consolidated these trends at a global scale, documenting that data centers consumed between 240 and 340 TWh in 2022 and projecting continued growth due to the expansion of AI. Wang et al. (2024) expanded the analysis to include macroeconomic variables, showing that the environmental impact of AI is non-linear and dependent on the composition of the regional energy matrix, which implies that identical computational systems can have radically different carbon footprints depending on their geographic context.

Green AI as a Theoretical and Methodological Framework

The concept of Green AI, systematized by Ferraz (2024), proposes that the evaluation of artificial intelligence systems should incorporate energy efficiency, environmental impact, and responsibility as first-order criteria, alongside traditional accuracy metrics — that is, the degree

to which a model correctly predicts outcomes — and speed. This framework rejects the separation between performance and sustainability, arguing that energy-efficient systems are, ultimately, systems with superior overall performance.

For the present research, Green AI serves two complementary functions: it provides the conceptual justification for focusing on data centers as the main point of intervention in the AI lifecycle, and it offers analytical tools to assess infrastructure practices through metrics such as PUE (Power Usage Effectiveness), carbon intensity of energy sources, and processing-unit efficiency. The suitability and limitations of these metrics become a central object of investigation throughout the analysis.

The Open Compute Project

The Open Compute Project was founded by Facebook (now Meta) in 2011, following the public release of the technical specifications for the data center built in Prineville, Oregon. The original premise was that hardware efficiency did not need to be a proprietary competitive advantage: by making the specifications public, the company aimed to accelerate innovation across the sector, reducing costs and improving energy efficiency through collaboration. Since then, OCP has grown into a foundation with over 400 member organizations, including major data center operators, hardware manufacturers, and technology companies (Drake, 2025).

Independent studies indicate that the OCP standard is perceived as a benchmark for efficiency in the industry. Research conducted by IHS Markit (2018) with European data center operators revealed that two-thirds of respondents considered OCP equipment as the main route to reducing energy consumption. This perception is reflected in market data: global spending on

infrastructure based on OCP specifications is expected to grow from USD 132 billion in 2025 to USD 295 billion in 2029, with a compound annual growth rate of 22.2% (Drake, 2025).

Efficiency Metrics and Their Limitations

PUE is the most widely used metric for assessing the energy efficiency of data centers. Defined as the ratio between the total energy consumed by the facility and the energy actually delivered to IT equipment, PUE indicates the proportion of energy wasted in supporting systems such as cooling and electrical distribution. A PUE of 1.0 would indicate perfect efficiency, while values below 1.5 are considered efficient by the industry (Ferraz, 2024).

Recent studies, however, document structural limitations of PUE as a sustainability indicator. Niewenhuis et al. (2024) showed that PUE ignores the source of the energy consumed, a critical factor for evaluating carbon emissions: depending on the energy mix, the difference in emissions between data centers with identical PUE can be up to twentyfold. Ligozat et al. (2022) highlighted that analyses limited to operational energy systematically underestimate environmental impact by disregarding hardware manufacturing, transport, and disposal. Sharma (2026), in a recent IEEE Spectrum publication, estimated that conventional metrics capture only about 30% of IT sector emissions, leaving out the largest portion associated with hardware production and end-user devices.

This body of evidence suggests that the widespread adoption of PUE, while representing progress compared to the absence of standardization, has created a specific methodological risk: by optimizing a partial metric, data center operators may achieve operational efficiency gains without necessarily reducing their total carbon footprint. This structural limitation forms the starting point for the original contribution of this research, developed in the discussion section.

Methodology

Case Study Design

This study adopts the case study as its central methodological strategy, as defined by Yin (1998), because it is suitable for investigating contemporary phenomena in real-world contexts where the researcher does not control events and seeks to understand how practices are actually implemented. The case study approach allows for capturing the complexity and contextuality of the relationships between technical innovation, organizational standardization, and environmental outcomes — dimensions that predominantly quantitative methods tend to simplify.

The design employed is a holistic single-case study of a descriptive nature, with OCP as the central unit of analysis. The choice of a single case is justified by its characteristics in the studied context, being an organization that combines technical openness with global-scale action on energy efficiency practices. In these terms, the case can be classified under Yin's (1998) typology as a single case, whose in-depth analysis allows for examining the relationships between practices, standards, and outcomes within its specific context.

To operationalize the investigation, three theoretical propositions were formulated: regarding the impact of AI growth on energy consumption (P1), the dependence of efficiency on combined technical and organizational solutions (P2), and the role of standardization in mitigating environmental impacts (P3). Additionally, three guiding hypotheses were established: H1, that the adoption of OCP-associated practices is related to lower energy consumption per processing unit; H2, that metric standardization contributes to greater transparency and comparability across operations; and H3, that sustainability outcomes result from the interaction

between guidelines, operational practices, and standardization, rather than from isolated technological solutions.

Evidence Collection Protocol

Following Yin's (1998) recommendations for ensuring reliability, the research developed a formal evidence collection protocol structured around seven analytical questions (Q1 to Q7), directly linked to the theoretical propositions. The guiding question is: how can data centers be developed and operated in an environmentally sustainable way, considering the high energy consumption associated with AI?

To investigate this, the following analytical questions were defined: Q1. What sustainability strategies are adopted in the development and operation of OCP data centers? Q2. How do these strategies manifest in the operational practices of data centers? Q3. How does the interaction between development principles, operational practices, and standardization influence the energy sustainability of data centers on a global scale? Q4. How do cooling strategies affect energy consumption throughout data center operations? Q5. To what extent does collaboration among OCP members contribute to the adoption of sustainable practices and innovations in energy efficiency? Q6. What challenges and limitations are observed in implementing these practices, and how do they impact energy consumption? Q7. Which energy performance indicators are most relevant for guiding decisions on the design, operation, and upgrading of data centers?

Evidence was collected from four categories of sources: academic publications, independent reports, technical and industry publications, and official documentation from OCP

and participating companies. For analysis, a reliability hierarchy was adopted based on two criteria: proximity to the phenomenon and potential for bias.

The gold standard corresponds to institutional documentation from OCP and its members, providing direct data but with higher risk of bias. The silver standard includes technical reports and specialized publications, with an intermediate position regarding specificity and independence. The bronze standard comprises academic publications and independent reports, characterized by lower bias but also lower proximity to the object of study. This structure does not establish a linear quality hierarchy but acknowledges the tension between data access and analytical independence.

A notable limitation concerns the difficulty in accessing direct primary sources. Yin (1998) emphasizes that interviews with key participants are among the most valuable evidence in case studies, as they allow researchers to obtain the actors' own interpretations of the phenomena. In this study, attempts were made to contact OCP members and representatives through two channels: via LinkedIn, with no response by the time of submission; and via institutional email, with responses still pending. This lack of direct access is recognized as a methodological limitation: it prevents verification through interviews and restricts the analysis to documentary and secondary sources.

Given this constraint, and in accordance with Yin (1998) — when the collection of evidence through one channel is obstructed, the researcher must compensate with greater rigor in triangulating the available sources — the study adopted a procedure of expanding and cross-checking documentary evidence to compensate for the absence of primary data from interviews. In practice, this meant: (a) systematically seeking sources of different types and

origins for each topic investigated, so that no relevant claim would be based on a single source; (b) comparing data from institutional sources, such as OCP documentation and reports from member companies, with data from independent sources, such as academic publications and reports from organizations like the IEA, identifying convergences and divergences; (c) organizing the evidence according to the degree of analytical independence, assigning greater weight to sources with lower risk of bias when contradictions arose; and (d) clearly recording, throughout the analysis, when a piece of information came solely from an institutional source with a stake in the subject, so that the reader can assess the weight of that evidence. This triangulation procedure does not eliminate the limitation resulting from the absence of interviews, but it allows the documentary analysis to be conducted with rigorous methodological control, increasing the internal validity of the findings.

Data Collection Process

Step 1 — The collection began with structured searches in publicly accessible search engines, mainly Google and Google Scholar. The searches were conducted in Portuguese and English, using terms such as "Open Compute Project energy efficiency," "data center sustainability," "PUE," "OpenBMC stranded power," "embodied carbon data centers," and "OCP Open Rack specifications," along with variations. The results were filtered by relevance and, when possible, by recency, prioritizing publications from 2019 onward. Institutional pages, articles from specialized media outlets, and technical repositories were included.

Step 2 — The official OCP website was analyzed in a structured manner to identify its public technical documentation. Sections covering specifications (Open Rack v2 and v3, Open Server, Open Network), project reports, event materials, sustainability content, and initiatives were examined. Documents available for download, such as technical reports and specifications,

were read and assessed for their relevance to the study questions. Open-source repositories linked to OCP were also consulted, including historical files on GitHub.

Step 3 — Artificial intelligence tools were used as support throughout the collection process. They were employed to locate articles, reports, and news on specific topics not captured in the initial searches, verify the existence of sources (such as URLs, authors, and titles), identify gaps in topic coverage, and compare information across different sources to guide the selection of materials for full reading. The outputs of these tools were treated only as a starting point. No information was used without direct verification in the original source.

Step 4 — Sources were selected based on two criteria: relevance to the study questions and reliability, assessed according to the adopted classification (gold, silver, and bronze standards). Duplicate sources were consolidated, and sources with conflicting data were retained for comparison in the analysis. The evidence used is listed in the references section, allowing verification and traceability of the information presented.

Analysis Strategy

The analysis follows a theoretical propositions–based strategy, as recommended by Yin (1998), in which collected evidence is examined in light of preformulated propositions. To organize the analysis, the data were structured into four dimensions: energy distribution, hardware design, software management, and circularity, representing different levels of intervention in data center operations.

Within each dimension, evidence is analyzed in relation to the analytical questions (Q1 to Q7) and the theoretical propositions, allowing identification of patterns of convergence and

divergence between observed practices and theoretical expectations. Results are then interpreted in an integrated manner, considering interactions between these dimensions and their relation to the reviewed literature.

Results

Layer 1: Reconfiguration of Power Distribution Architecture

The starting point for understanding OCP is to examine how electricity flows through a traditional data center and where it is lost along the way.

In a conventional data center, high-voltage electricity from the grid passes through multiple stages before reaching a server's processor. First, a transformer reduces the voltage to a level suitable for building use. Next, power is distributed to racks, which organize servers and other IT equipment, via Power Distribution Units (PDUs) that direct electricity to individual devices. Each server then receives power through a Power Supply Unit (PSU), which converts electricity to levels appropriate for internal components. Finally, energy is adjusted and distributed within the server to processors and other circuits.

Each stage dissipates some energy, typically as heat. Typical losses include about 2% at the transformer, 6–12% in the backup system (UPS, which keeps servers running during power outages), 3% in the PDU, and up to 10% in the PSU, totaling up to 27% along the full path. The largest losses occur in PSUs and backup systems, especially in higher-redundancy data centers such as Tier III, which can consume up to 1.78 times more energy than Tier II facilities. Literature highlights that reducing PSU losses is a key intervention point to improve energy efficiency (Wiboonrat, 2021).

OCP proposed changes at three specific points in this chain. First, the elimination of a transformation stage: while traditional data centers reduce grid voltage to 208 V before reaching equipment, OCP delivers power directly from the transformer to the rack at 277/480 V (or 230/400 V in Europe), achieving approximately 2.5% efficiency gain (Gero, 2018). Second, the migration from 12V to 48V: traditionally, PSUs supply 12 volts DC to servers. In the OCP standard, with Open Rack v2 and v3, power is distributed at 48 volts DC. According to Joule's law, energy loss along a conductor increases with the square of the current. By increasing voltage from 12V to 48V, current is reduced fourfold, lowering heating losses in cables. This allows thinner cables and changes the thermal profile of the racks (Electronic Specifier, 2020). Google reported a 30% increase in energy efficiency using this standard, although the disclosure occurred in an institutional context favorable to OCP, with the physical gain being independent (Zipfel, 2016). Third, centralization of power supplies at the rack level: in OCP, instead of each server having its own PSU, power is concentrated in a power shelf that supplies the entire rack. AC-to-DC conversion occurs only once, eliminating losses at each server. Servers connect to the rack's power bar through a blind-mate connector, removing the need for internal PSUs, reducing costs, and lowering heat generation. This design also eliminates internal rack cabling, simplifying installation and maintenance (Super Micro Computer, 2024).

From a dissemination perspective, Open Rack specifications are publicly available in the foundation's repository and can be adopted by any operator. A survey by IHS Markit (2018) with European operators found that two-thirds considered OCP equipment as a pathway to reducing energy consumption, suggesting that perceptions of technical efficiency can influence adoption decisions.

Layer 2: Functionality-Oriented Design (Vanity-Free)

The second layer of intervention operates at the physical level of racks and servers, challenging geometric constraints inherited from a century-old standard. The 19-inch rack format was created by AT&T around 1922 to organize telephone exchange equipment and was later formalized by the EIA-310 specification for the IT industry (Nelson, 2016). As a result, generations of servers were designed according to a standard that imposes geometric restrictions limiting thermal efficiency: bezels (front panels or external protective frames), front panels with logos, and aesthetic frames that serve brand differentiation rather than technical performance (Hall, 2022).

OCP proposed the Open Rack standard, based on the vanity-free design principle: all surfaces and components must serve a technical function, while purely aesthetic elements are eliminated. The energy impact of this decision operates through two interdependent mechanisms.

The first is aerodynamic: bezels and front panels act as obstacles to airflow through the racks. ScaleUp Technologies, a European operator that adopted OCP hardware and documented results in a real operational environment, recorded energy consumption reductions of 15% to 20% compared to conventional 19-inch servers, with OCP servers consuming even less energy under low-load conditions (Streit, 2021). It should be noted that ScaleUp Technologies is a silver-standard source — a commercial operator with an institutional interest in demonstrating the benefits of its technological choice.

The second mechanism is geometric. The Open Rack has an internal width of 537 mm (21 inches), compared to 19 inches in the EIA-310 standard, while maintaining an external width of 600 mm for compatibility with existing installations. This difference allows the use of larger-diameter fans — 80 mm in OCP servers versus 40 mm in conventional servers, as

documented by ScaleUp Technologies (2021) — which move the same volume of air at lower rotational speeds and therefore consume less energy. Additionally, the standard rack unit measurement (OpenU = 48 mm, versus 44.45 mm in conventional racks) accumulates space across multiple rack slots, improving airflow between components (Sarti & Mills, 2012).

The energy context amplifies the relevance of these interventions. The IEA (2025) estimates that cooling accounts for between 7% and more than 30% of a facility's total energy consumption, depending on efficiency levels. In less efficient conventional data centers, this proportion can reach 38% to 40% (Offutt & Zhu, 2024). A global survey by Supermicro with 1,362 operators indicated that roughly 12% of facilities meet efficiency criteria, with less efficient cooling systems being a recurring factor in overall consumption (Anderson, 2019).

A relevant limitation must be noted: migrating conventional infrastructure to the Open Rack standard involves non-trivial transition costs, as pre-installed 19-inch racks are not directly compatible with OCP servers. For smaller operators, this cost can outweigh energy savings over reasonable time horizons — an aspect still poorly quantified in the literature. Furthermore, the model introduces an ecosystem dependency: efficiency gains are contingent on integrated adoption of the OCP ecosystem (rack, power supplies, and management software), which raises entry barriers and limits applicability in hybrid infrastructures.

Layer 3: Open Energy Management Software

The third layer of intervention addresses energy control and monitoring at the server and rack level. Each server is equipped with a BMC (Baseboard Management Controller), a dedicated microprocessor that monitors temperature, voltage, and power consumption independently of the main processor. The BMC operates through firmware, embedded software

that defines its hardware management and control. Historically, this firmware was proprietary, varying across manufacturers and based on the IPMI (Intelligent Platform Management Interface) protocol, which has security and energy management precision limitations (Frazelle, 2020). The diversity of proprietary solutions made standardized, granular energy management in large-scale environments difficult.

In response, engineers at Meta developed an open-source BMC firmware prototype in 2014, made public in 2015, and migrated in 2018 to the Linux Foundation as a collaborative project, OpenBMC, with participation from Microsoft, Intel, IBM, Google, and Meta (Fang, 2015; Zemlin, 2018). Being Linux-based and open-source, OpenBMC can be audited, corrected, and enhanced by multiple organizations. Independent evidence indicates that this openness has a positive impact on both security and energy monitoring capabilities (Larabel, 2018).

Building on OpenBMC, OCP developed OpenRMC (Open Rack Management Controller), which extends management from individual servers to the entire rack. The primary structural problem OpenRMC seeks to address is stranded power: in traditional data centers, operators allocate more electrical capacity than servers actually demand to ensure a safety margin during peak loads. When this margin is oversized, the data center operates with installed capacity significantly higher than actual consumption (Tozzi, 2025). This waste is exacerbated by the fact that servers consume between 30% and 60% of their maximum power even at low utilization (NRDC, 2014).

However, adopting OpenBMC involves significant practical limitations. As documented by Redinent (2025), its implementation requires specialized expertise in embedded Linux firmware, shifting the cost from proprietary licensing to the need for skilled technical teams.

Large operators can absorb this cost within existing structures, but for smaller organizations, it can represent a substantial barrier. There is also a governance dimension: the Linux Foundation is predominantly composed of large technology companies that benefit from the standardization promoted by the project, meaning development priorities tend to reflect the interests of these dominant actors.

Layer 4: Circularity and Life Cycle Design

The fourth layer addresses a dimension that has been largely neglected in sustainability studies of digital infrastructure: the energy consumed before a data center becomes operational and the impact of design decisions on equipment lifespan.

To contextualize this concept, it is useful to refer to the GHG Protocol (World Resources Institute & WBCSD, 2004), the leading international framework for emissions accounting. Scope 1 emissions are direct (e.g., diesel generators), scope 2 are indirect emissions associated with electricity consumption, and scope 3 encompasses all other emissions across the value chain: server manufacturing, equipment transportation, building construction, and hardware disposal. The iMasons Climate Accord (2024), an independent organization focused on decarbonizing digital infrastructure, documents that scope 3 emissions can represent between 38% and 69% of a data center's total carbon footprint over its lifetime. While this range still requires validation through independent studies with comparable methodologies, it clearly indicates that operational efficiency gains capture only a portion of environmental impact.

In conventional data centers, this problem is amplified by incompatibility between equipment from different manufacturers and generations, which often forces the replacement of entire racks when only some components are obsolete. The modular design proposed by OCP

seeks to mitigate this effect by standardizing components with open, interchangeable specifications, allowing selective replacement of system parts. Schneider Electric (2023, cited in Lin, 2023) estimates that extending server lifespans by one year can reduce cumulative embodied carbon by roughly 16%, though this decision must be weighed against the energy efficiency gains of newer equipment.

To address the lack of data on embodied emissions, OCP and the iMasons Climate Accord published in 2024 the Embodied Carbon Disclosure — OCP Base Specification, a framework for standardizing the reporting of embodied carbon. The initiative represents a transparency advance but is recognized as an initial step: it defines reporting categories without establishing standardized calculation methodologies or independent verification mechanisms, which may result in data that is not directly comparable across suppliers.

The circularity approach also faces economic and market limitations. The absence of a structured secondary market for OCP components reduces the practical viability of reuse: for modularity to translate into effective circularity, replaced components must find new uses, which remains underdeveloped in the OCP ecosystem, whose adoption is still concentrated among large operators (iMasons Climate Accord, 2024). Additionally, there is a tension between modularity and technological innovation: interchangeability presumes cross-generational compatibility, but the rapid evolution of AI architectures introduces changes in formats, interfaces, and thermal requirements that hinder backward compatibility.

Data from Clarity AI (Fettes, 2025) indicate that, among the eight largest global data center operators, only one discloses quantified reuse targets at the fleet scale, reflecting low practical institutionalization of circularity even where technical capacity exists.

Discussion

Verification of Theoretical Propositions

The results allow for verification of the three theoretical propositions formulated in the study protocol. Proposition P1, which states that the growing demand for AI drives increased energy consumption in data centers, is supported by documented consumption data from the IEA (2023, 2025) and market growth projections reported by Drake (2025). Structural growth driven by continuous inference workloads at scale highlights the relevance of the interventions analyzed.

Proposition P2, which posits that energy efficiency in data centers depends on both technical solutions and organizational processes, is consistently observed across the four layers. In none of the layers do efficiency gains arise solely from technical innovation: Layer 1 relies on open standardization for dissemination; Layer 2 on the elimination of historical industrial conventions; Layer 3 on open-source code and interorganizational collaboration; and Layer 4 on the development of disclosure frameworks and the presence of secondary markets. The evidence indicates that technical and organizational factors are interdependent, as technical innovation alone cannot produce efficient practices, and without open standardization and collaborative governance, these practices cannot scale globally.

Proposition P3, which suggests that integration of design, operation, and metric standardization can reduce environmental impacts, is observed but with an important qualification: results show that even when integration is structured from a technical-operational perspective, it remains limited if the adopted metrics do not account for the full life cycle. Wang et al. (2024) indicate that AI's environmental impact is non-linear and depends on the regional

energy mix, meaning strategies such as hardware reuse produce varying effects depending on the energy context. The systemic integration proposed by P3 is therefore necessary but not sufficient and must be accompanied by appropriate metrics for results to be accurately assessed.

Original Contribution: The Structural Inadequacy of PUE as a Sustainability Indicator

The primary finding of this research emerges from the integrated analysis of the four layers: PUE, a metric commonly used to assess data center efficiency, has limitations in guiding environmental sustainability decisions, with implications for sector strategy and operations.

The analysis identifies three dimensions of this inadequacy. At the internal operational level (Layer 3), PUE does not capture inefficiencies within equipment, such as energy allocated preemptively to underutilized components (stranded power). A data center may report a PUE of 1.07 and still waste significant energy in underutilized servers, an inefficiency invisible to the metric.

At the energy source dimension (Layer 3), Niewenhuis et al. (2024) demonstrated that two data centers with identical PUE can have carbon footprints differing by up to twentyfold, depending on the energy mix. A solar-powered data center with a PUE of 1.5 has a much lower environmental impact than an equivalent coal-powered facility with a PUE of 1.2, yet the metric would not reflect this difference.

At the lifecycle dimension (Layer 4), PUE measures only post-commissioning performance, excluding scope 3 emissions, which, according to the iMasons Climate Accord (2024), account for 38% to 69% of the total footprint. Ligozat et al. (2022) indicate that analyses

restricted to operations tend to underestimate environmental impact, and Sharma (2026) estimates that conventional metrics capture roughly 30% of total IT sector emissions.

These three dimensions constitute what this research terms the partial metric paradox: optimizing PUE can yield operational gains while ignoring a large portion of the environmental footprint. This means that improvements reported in PUE may be technically correct but do not fully reflect data center sustainability.

The contribution of this research is to show that PUE's insufficiency is a scope limitation, not a marginal technical issue, and that adequate sustainability assessment requires a more comprehensive framework. The data suggest articulating at least three categories of indicators: (a) operational metrics adjusted for energy mix, such as CUE (Carbon Usage Effectiveness); (b) computational intensity metrics, relating performance to energy consumption; and (c) lifecycle metrics, incorporating scope 3 emissions with standardized methodologies. The Embodied Carbon Disclosure Specification published by OCP and the iMasons Climate Accord (2024) represents a step toward this integration, though it still lacks comparable methodologies across manufacturers and independent verification mechanisms.

The Paradox of Open Governance

A second relevant dimension emerged from the analysis: the relationship between technical openness and institutional concentration within the OCP ecosystem. The open standardization model, in which technical specifications are published freely and can be adopted, adapted, and improved by any organization, facilitates the dissemination of energy efficiency practices. The extent of this reach is evidenced by adoption data from IHS Markit (2018) and market growth projections (Drake, 2025).

However, the analysis of Layers 3 and 4 shows that this openness operates in a non-neutral institutional environment. The Linux Foundation (responsible for OpenBMC) and the iMasons Climate Accord (responsible for the Embodied Carbon Disclosure Specification) are predominantly composed of the same large technology companies that are also the main beneficiaries of the practices being standardized. This creates a dynamic in which the actors with the greatest power to define sustainability criteria are precisely those with the highest technical capacity to meet them, while smaller operators face implementation barriers that technical openness alone does not resolve (Redinent, 2025).

Verhaeghe (2025) argues that mitigating the energy impacts of computing depends on changes in technological governance models, not just technical innovations. The findings of this research support this argument: code openness is a necessary condition, but not sufficient, for democratizing energy efficiency gains. Systemic sustainability additionally requires governance mechanisms that expand access to technical knowledge and lower implementation barriers for organizations with less installed capacity.

Study Limitations

This study has several limitations that should be considered when interpreting the results. The main limitation is the lack of direct access to primary sources through interviews: attempts to contact OCP representatives did not receive responses by the time of submission, restricting the analysis to documentary and secondary sources. Yin (1998) emphasizes that interviews with key participants are among the most valuable sources of evidence in case studies, as they provide the actors' own interpretations of the phenomena under investigation.

Additionally, the single-case design, while justified by the specific nature of OCP, limits the generalizability of the findings. Results cannot be directly applied to other open-standard contexts without considering the particular conditions of each case. The intended generalization is theoretical, as Yin (1998) recommends, rather than statistical: the findings contribute to formulating propositions about the relationship between open standardization, governance, and energy sustainability, but do not allow inferences about populations of data centers.

Conclusion

The analysis indicates that energy sustainability in data centers depends on the coordinated combination of multiple interventions — in power distribution architecture, hardware design, software management, and lifecycle strategies — rather than on isolated solutions. Existing interventions show significant potential to reduce energy consumption and associated emissions, but the actual environmental impact remains limited when metrics such as PUE do not fully capture all relevant aspects, including energy source and embedded hardware emissions.

Furthermore, the adoption of efficient solutions faces economic, technical, and structural barriers, and the lack of broad standardization and regulatory incentives limits the dissemination of best practices. In this context, the central challenge is not the absence of technology but how sustainability is measured, implemented, and incentivized.

The study suggests that effective advances require integration of technical innovation, governance, comprehensive metrics, and operational models that account for the full system lifecycle. The Open Compute Project (OCP) exemplifies this potential by combining efficient

design, open standardization, and replicable operational practices, even though its solutions do not fully cover all dimensions of sustainability.

References

Electronic Specifier. (2020). *Advanced Energy announces Open Rack version 3 power shelf to support Open Compute Project and evolution to 48-volt infrastructure*. Electronic Specifier. https://www.electronicspecifier.com/products/power/open-rack-version-3-power-shelf-to-support-open-compute-project/

Anderson, M. (2019). How green was my data center? (Not very). *IEEE Spectrum*. https://spectrum.ieee.org/how-green-was-my-data-center-not-very

Drake, C. (2025). *IDC market impact study: OCP Global Summit 2025*. IDC. https://146a55aca6f00848c565-a7635525d40ac1c70300198708936b4e.ssl.cf1.rackcdn.com/images/2538f9e7afe4473fb199c4f688b9d954a101db6a.pdf

Fang, T. (2015, 10 de março). Introducing "OpenBMC": An open software framework for next-generation system management. *Engineering at Meta*. https://engineering.fb.com/2015/03/10/open-source/introducing-openbmc-an-open-software-framework-for-next-generation-system-management/

Ferraz, I. (2024). Green AI: Metrics for energy efficiency and sustainability of artificial intelligence algorithms and data centers. *Brazilian Journal of Environmental and Sustainability Management, 11*(29). https://revista.ecogestaobrasil.net/v11n29/v11n29a14a.html

Fettes, N. (2025). Data center emissions are rising. Are firms doing enough? *Clarity AI*. https://clarity.ai/research-and-insights/climate/data-center-emissions-are-rising-are-firms-doing-enough/

Frazelle, J. (2020). Opening up the baseboard management controller. *Communications of the ACM, 63*(2), 38–40. https://doi.org/10.1145/3373376

Gero, K. (2018, 21 de março). Applying Open Compute rack and power specs in your data center. *Data Center Knowledge*. https://www.datacenterknowledge.com/data-center-chips/applying-open-compute-rack-and-power-specs-in-your-data-center

Hall, C. (2022, 3 de junho). Open Compute Project: Data center design by community effort. *Data Center Knowledge*. https://www.datacenterknowledge.com/data-center-chips/open-compute-project-data-center-design-by-community-effort

IHS Markit, Technology Group. (2018). *European data center energy-efficiency study*. https://f990335bdbb4aebc3131-b23f11c2c6da826ceb51b46551bfafdc.ssl.cf2.rackcdn.com/images/c8c5d0b593fb4519659087527669bd418296379b.pdf

iMasons Climate Accord. (2024, 23 de abril). Open Compute Project Foundation and iMasons announce a new collaboration. https://climateaccord.org/press-releases/open-compute-project-foundation-and-imasons-announce-a-new-collaboration/

International Energy Agency. (2023, 11 de julho). *Data centres and data transmission networks*. https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks

International Energy Agency. (2025). *Energy and AI*. https://www.iea.org/reports/energy-and-ai

Larabel, M. (2018, 19 de março). Linux Foundation announces OpenBMC project to create open-source BMC firmware. *Phoronix*. https://www.phoronix.com/news/Linux-Foundation-OpenBMC

Li, P., Liu, Y., Yang, J., & Ren, S. (2024). Towards socially and environmentally responsible AI. *arXiv*. https://doi.org/10.48550/arXiv.2407.05176

Ligozat, A.-L., Lefevre, J., Bugeau, A., & Combaz, J. (2022). Unraveling the hidden environmental impacts of AI solutions for environment: Life cycle assessment of AI solutions. *Sustainability, 14*(9), 5172. https://doi.org/10.3390/su14095172

Lin, P. (2023, 11 de julho). Demystifying data center Scope 3 carbon with our findings. *Schneider Electric Blog*. https://blog.se.com/datacenter/2023/07/11/demystifying-data-center-scope-3-carbon-with-our-findings/

Luccioni, A. S., Viguier, S., & Ligozat, A.-L. (2022). Estimating the carbon footprint of BLOOM, a 176B parameter language model (arXiv:2211.02001). *arXiv*. https://doi.org/10.48550/arXiv.2211.02001

Natural Resources Defense Council. (2014, agosto). *Data center efficiency assessment: Scaling up energy efficiency across the data center industry* (Issue Paper IP:14-08-a). https://www.nrdc.org/sites/default/files/data-center-efficiency-assessment-IP.pdf

Nelson, R. (2016). Why is the 19-inch rack 19 inches, or is it? *Electronic Design*. https://www.electronicdesign.com/technologies/test-measurement/article/21206542/why-is-the-19-inch-rack-19-inches-or-is-it

Niewenhuis, D., Talluri, S., Iosup, A., & De Matteis, T. (2024). FootPrinter: Quantifying data center carbon footprint. In *Companion of the 15th ACM/SPEC International Conference on Performance Engineering* (ICPE '24 Companion). ACM. https://doi.org/10.1145/3629527.3651419

Offutt, M. C., & Zhu, L. (2024). *Data centers and energy: Congressional issues* (CRS Report No. R48646). Congressional Research Service. https://www.congress.gov/crs_external_products/R/PDF/R48646/R48646.1.pdf

Redinent. (2025). OpenBMC adoption in datacenter fleet. https://global.redinent.com/global/openbmc-adoption-in-datacenter-fleet/

Sarti, P., & Mills, S. (2012). *Open Compute Project: Open Rack hardware v1.0*. Open Compute Project. https://github.com/facebookarchive/opencompute/blob/master/open_rack/spec/Open_Compute_Project_Open_Rack_v1.0.pdf

Schafer, I. C. (2018, 19 de março). OpenBMC and Linux Foundation join forces to codify open-source BMC firmware stack. *ITOps Times*.

https://itopstimes.com/cloud/openbmc-linux-foundation-join-forces-codify-open-source-bmc-firmware-stack/

Schneider, I., Xu, H., Benecke, S., Patterson, D., Huang, K., Ranganathan, P., & Elsworth, C. (2025). Life-cycle emissions of AI hardware: A cradle-to-grave approach and generational trends. *arXiv*. https://arxiv.org/pdf/2502.01671v1

Sharma, A. (2026). Data center sustainability metrics: Hidden emissions—we're measuring data center sustainability wrong. *IEEE Spectrum*. https://spectrum.ieee.org/data-center-sustainability-metrics

Smolaks, M. (2016, 5 de agosto). Google and Facebook unleash the Open Rack standard. *Data Center Dynamics*. https://www.datacenterdynamics.com/en/news/google-and-facebook-unleash-the-open-rack-standard/

Streit, C. (2021, 23 de abril). Specifics of OpenCompute servers: How OCP hardware differs from classic hardware. *ScaleUp Technologies*. https://www.scaleuptech.com/en/blog/differences-open-compute-server-and-standard-hardware/

Strubell, E., Ganesh, A., & McCallum, A. (2019). Energy and policy considerations for deep learning in NLP. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pp. 3645–3650. https://doi.org/10.18653/v1/P19-1355

Super Micro Computer, Inc. (2024, abril). *Supermicro total OCP rack solution* [Solution brief]. https://www.supermicro.com/solutions/Solution-Brief-Supermicro-OCP.pdf

Tozzi, C. (2025, 13 de outubro). Stranded power: The hidden challenge reshaping data center energy strategy. *Data Center Knowledge*. https://www.datacenterknowledge.com/energy-power-supply/stranded-power-the-hidden-challenge-reshaping-data-center-energy-strategy

Verhaeghe, P. (2025). *Rethinking digital infrastructure governance for a sustainable AI ecosystem*. Recuperado de publicação setorial especializada sobre governança de infraestrutura digital.

Wang, Q., Li, Y., & Li, R. (2024). Ecological footprints, carbon emissions, and energy transitions: The impact of artificial intelligence (AI). *Humanities and Social Sciences Communications, 11*, 1043. https://doi.org/10.1057/s41599-024-03520-5

Wiboonrat, M. (2021). Energy loss model for Tier 2 and Tier 3 data centers. *Suranaree Journal of Science and Technology, 29*(1), 010092. https://www.thaiscience.info/Journals/Article/SJST/10994321.pdf

World Resources Institute & World Business Council for Sustainable Development. (2004). *The greenhouse gas protocol: A corporate accounting and reporting standard* (Rev. ed.). https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf

Wu, C.-J., et al. (2021). Sustainable AI: Environmental implications, challenges and opportunities. *arXiv*. https://arxiv.org/pdf/2111.00364

Yin, R. K. (1998). *Case study research: Design and methods* (2ª ed.). Sage Publications.

Zemlin, J. (2018, 19 de março). OpenBMC project community comes together at the Linux Foundation to define open source implementation of BMC firmware stack. *The Linux Foundation*. https://www.linuxfoundation.jp/blog/2018/03/openbmc-project-community-comes-together-at-the-linux-foundation-to-define-open-source-implementation-of-bmc-firmware-stack/

Zipfel, J. (2016, March 9). Google joins Open Compute Project to drive standards in IT infrastructure. Google Cloud Blog. https://cloud.google.com/blog/products/compute/google-joins-open-compute-project-to-drive-standards-in-it-infrastructure